

Eventive Carpooling – Final Year Project Summary

****Project Title:**** Eventive Carpooling – A Route Propagation Based Dynamic Carpooling System

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Abstract

Eventive Carpooling is a smart mobility solution designed to address urban traffic congestion, transportation costs, and environmental concerns through a flexible, route-aware carpooling system. Unlike conventional carpooling platforms that match riders with identical destinations, our system implements dynamic route propagation—allowing drivers to accept new passengers along their ongoing route, essentially enabling a ride that never ends.

This intelligent system supports multi-hop trips where a passenger can reach a destination using a chain of rides, based on real-time availability. Users can plan, book, and pay for rides through a mobile application built with React Native, powered by Node.js, Firebase, and machine learning-based route matching. Safety features such as gender-specific ride options and live tracking are included to build user trust and broaden adoption.

Key Features

- Dynamic route propagation — continuous matching of riders with drivers along the route.
- Multi-hop ride chaining — passengers can complete journeys using a sequence of compatible rides.
- Live location tracking via Google Maps API.
- Gender-specific carpooling for enhanced safety.
- Ride cost sharing, route optimization, and environmental benefits.
- Built-in event creation: drivers can schedule rides with specific time, destination, and seat availability.

Technology Stack

- Frontend: React Native (for Android)
- Backend: Node.js, Firebase (Authentication & Realtime DB)
- Database: MongoDB

- Machine Learning: Route similarity and fare prediction
- APIs: Google Maps API (route mapping), Online Payment Integration

Problem Statement

Most current carpooling solutions only connect drivers and riders with exactly matching origins and destinations. This significantly limits the potential of ride-sharing. Moreover, female users often feel insecure using mixed-gender ride-sharing services. Our platform solves both challenges:

- By enabling non-identical route-based matching using intelligent algorithms.
- By offering gender-aware pooling options to make travel safer and more inclusive.

Implementation Highlights

- Route Matching Algorithm: Combines Google Maps routing with Dijkstra's algorithm to identify optimal ride sequences.
- Heuristic and Euclidean Distance Methods: For fast and efficient location matching.
- Ride Event Casting: Drivers can create ride events, the system evaluates passenger interest in real-time.
- Safety First: Firebase authentication, ride tracking, and gender-filtered ride options are incorporated.

System Architecture Overview

- Mobile App: Interface for passengers and drivers.
- Admin Portal (Web): Admin-level monitoring and management.
- Core Engine: Logic for ride propagation, chaining, and ML-based suggestions.

Benefits

- Cuts down fuel usage and cost by enabling shared rides.
- Reduces city congestion by decreasing single-occupancy vehicle usage.
- Promotes a cleaner environment through fewer emissions.
- Encourages community engagement by connecting users.
- Provides a safe, affordable, and scalable transport solution.

Evaluation and Testing

The system was tested for:

- Functionality: Real-time ride creation, matching, and payment workflow.
- Usability: Smooth UI/UX and minimal onboarding steps.
- Compatibility: Working on major Android versions and across devices.
- Performance: Stress-tested under high user load.
- Security: Data encryption and secure authentication.
- Feedback Loop: User feedback used for continual improvements.

Conclusion

Eventive Carpooling introduces an innovative shift in ride-sharing paradigms by integrating continuous ride propagation, safety-aware matching, and dynamic routing intelligence. It

redefines urban commuting with a focus on sustainability, accessibility, and user-centric design. The project demonstrates technical feasibility and real-world applicability, especially in urban areas facing transit bottlenecks.



National University
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FINAL YEAR PROJECT PROPOSAL

BS-CS FALL 2021

EVENTIVE

(Carpooling System)

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Project Details

1.1. Project Area of Specialization

App Development and ML.

1.2. Project Timeline

Project Start Date	Project End Date
September 2022	April 2023

1.3. Project Summary

Carpooling is one of the latest technologies discovered which has made travelling convenient and efficient to the common man. It also known as car-sharing in which one can travel to their destination while sharing the vehicle and the expenses incurred hence reduce single individuals travelling cost, fuel cost and tiredness. Due to the growth in the population, there is inadequate transportation through personal vehicles which gives rise to traffic congestion on roads and damages the eco-friendly environment by increasing air and noise pollution. Hence carpooling is the optimal solution to solve these problems. we will make an Android based application that will allow passengers to collaborate with other like-minded people and plan out their journey using the easy UI of the app after signing in to it.

The main aim is to provide a type of mobility service that, through a joint digital channel enables users to plan, book, and pay for multiple mobility services through carpooling. The concept describes a shift away from personally-owned modes of transportation and towards mobility provided as a service. The key concept is to offer travelers run time solutions based on their travel needs. This carpooling system shows that the user can get from one destination to another by using a combination of multiple rides. For e.g., if a user wishes to move from a point A to point C but fails to find a ride connecting the two points directly, our system will generate a combination of rides operating on that route, which the user can book simultaneously in order to reach its destination through a combination of carpools. In addition, the user can choose their preferred trip based on cost, time, and convenience. At that point, any necessary bookings would be performed as a unit. It is expected that this service should allow roaming, that is, the same end-user app should work in different cities. Furthermore, an individual's safety is also an important aspect in this scenario therefore this system provides gender specific carpooling and live tracking system for rides. Together with other emerging vehicular technologies such as connected cars and electric vehicles, carpooling is contributing to a new type of future mobility, which is autonomous, connected and shared.

1.4. Project Objectives

This Carpooling system aims to provide one stop solution to daily commute problem without any hassle, making it more flexible and reliable while creating great connections with the help of an e-platform where commuters come together to share pleasant journey. Carpooling is a revolutionary and a unique way of travelling. Whether you own a bike, car or a rider, just post your ride details in our app and we will

match you with co-riders on your way. We want to provide a hassle-free transit to our users along with a great experience.

- Enable rider to create events that would specify the following information:
 - The total vacancy in the car.
 - The time at which the event is going to take place.
 - The Final destination.
- Development of the logic that would enable-
 - Poll in the location information of all the intended recipients.
 - Take decision based on the context on the location.
 - Send SMS to all the selected recipients and handle the accepted or rejected messages received from the recipients.
- Generate a tracking system using Google Map that shows the initiator the map between his location, all the recipients that agreed to his car-pooling event and the final destination.
- Reduce overall traffic congestion on the roads and single occupancy car trips by implementing carpooling system.
- Promoting alternative modes of transport.
- Reduce peak hour congestion, pollution and other poisonous gases emission in air.
- Cost effective transit, by sharing the cost of driving one car.
- Reduces driving-related stress for participants
- Provide social connections in the society.

1.5 Literature Review / Background Study

Carpoolyn [1] is Pakistan's 1st Carpooling (Ride-Sharing) App with an aim to make Traveling accessible, affordable, and safe for Everyone Across Pakistan. It connects the drivers with passengers traveling on the same routes to share the fuel cost.

Have A Car? Just charge for empty seats in your car and make money to cover your commuting cost. Need A Ride? Share a ride with drivers heading towards the same destinations & reduce your traveling cost.

Ridely [2] is rideshare services in Pakistan. It's an emerging app that bring a new concept of route matching in Carpool. This carpooling app allows its users to select rides based on the route match between two cities of Pakistan.

Savaree [3] is a Pakistan-based cab service that offers affordable and safe rides. It offers a carpooling app that allows users to find rides to places they would like to go. It also matches passengers with drivers going along specific routes or to specific events. Savaree was founded in 2014 and is based in Lahore, Pakistan. It also provides inter-city services.

BlaBlaCar [4] is the world's largest long-distance ridesharing community. Conceived in December 2003 by Frédéric Mazzella, and founded in 2006, BlaBlaCar connects drivers and passengers willing to travel together between cities and share the cost of the journey. BlaBlaCar has more than 20 million members across 19 countries. Members must register and create a personal online profile, which includes ratings and reviews by other members, social members show how much experience they have of the service, meaning those with more-known as "ambassadors" - attract more ride shares.

FolksVagn [5] offers a community-based system that helps people share rides with others. While the passengers get rides at costs much cheaper than a regular taxi service, the car owner gets a share of the fare. It is open only to corporate clients as it requires a corporate email for registration and has a prepaid account or online wallet system to pay for the ride.

Taxi for sure [6], the famous taxi-hire application on android platform is the first car sharing application who took the initiative and introduced Carpooling for “Vacationers” i.e. for those who are on vacations and want to spend less on travelling to save their pocket.

Problem:

One major shortcoming of these applications is that all the applications are focusing on connecting drivers and passengers that have same destination. No application is focusing on connecting riders with dissimilar destination. Our main focus is to solve this issue by using efficient ML algorithms for real-time route matching.

Security is another concern specially for women. In today’s world it’s hard to rely on anyone. In our project we are focusing on gender specific mechanism of carpooling. In this way, we can create more opportunities for female drivers and passengers.

1.6. Project Implementation

Carpooling system provides the benefits of sharing rides with different peoples. By surveying different papers, it is found that it uses heuristic searching techniques for searching the cars. By using Euclidean distance techniques, it finds the nearest car that has to be assigned to passenger. It usually uses Global Positioning System (GPS) for finding the locations of cars. In study it is found that, Route matching techniques are used in order to match drivers and passenger’s locations by using Google maps API, and then based on their route, driver is allocated to passenger. In most of carpooling system, it is observed that the Dijkstra algorithm is used for detecting the shortest route of destination of passenger. Dijkstra algorithm is usually used for detecting the optimal minimum path. It is found that it checks for maximum possibilities and then chooses the minimum route for the journey. Similarly, Dijkstra algorithm is also used to connect nearest vehicle.

1.7. Benefits of Project

- **Save money:** Share the costs of driving with other riders.
- **Reduce stress:** Read, listen to music, or relax when you’re not driving.
- **Save time:** As a carpool, you can drive in the HOV (high occupancy vehicle) lane.
- **Help the environment:** Carpooling creates cleaner air and safer communities.
- **Prolong the life of your vehicle:** Shared driving puts fewer miles on your own car.

- **Reduces traffic:** When three people carpool, there are two fewer cars on the road. Automatically traffic on roads reduces.
- **Safety:** Our gender specific approach help female to travel more freely and securely.
- Employer financial tax incentives for supporting carpooling.

1.8. Technical Details of Final Deliverable

- Key Frame Extraction algorithms for CCTV videos streams
- Anomaly Detection of various suspicious activities.
- Aggregation of Anomalous Video Segmentation.
- UI Design so that multiple online/offline streams can be analyzed.

1.9. Final Deliverable of The Project

- An application that provides one stop solution to daily commute problem without any hassle, making it more flexible and reliable while creating great connections with the help of an e-platform where commuters come together to share pleasant journey.

1.10. Core Technology

- React Native, mongoDB, Nodejs, Machine Learning Models, Python.

References

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7. Systems (HPCC/SmartCity/DSS), 1271--1278, (2018)

Project Key Milestones

Elapsed time in (days or weeks or month or quarter) since start of the project	Milestone	Deliverable
Month 1	Practical Analysis of Existing System.	Literature review. Nature of the problem Identify possible challenges. analyze, models, and a logical alternative.
Month 2	Requirements Gathering.	Identify and gather insights of existing systems and get an adequate grasp of the project flow.
Month 3	Proposed Architecture.	Understanding the course of the project. ER Diagram, Data flow diagram, Activity Diagram etc.
Month 4	Prototype 1 + Alpha Testing.	Implementation of components i.e. of user and Driver portals (User login + Driver Registration). Casting Mechanism such as Any Cast. Geofencing.
Month 5	App Development and Promotion Website Development.	Interface development
Month 6	Applying ML/DL Algorithms.	Route Matching, fare calculator, recommendation based on rating, Radius calculator.
Month 7	Integration of Relevant APIs and Modules + Component Testing.	Google map API, Payment API.
Month 8	Testing.	System Testing.



Eventive Carpooling

PROJECT SUPERVISOR

Mr. Nouman Durrani

PROJECT CO-SUPERVISOR

-

PROJECT TEAM

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Muhammad Faiz	K190247
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Submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in
Computer Science.

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May 2023

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Submission Date	May 5, 2023	

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Acknowledgement

We would like to express our sincere gratitude to our FYP supervisor, Mr Nouman Durrani for their guidance, support, and encouragement throughout our project. Their insightful feedback, valuable suggestions, and constructive criticism have been instrumental in shaping our project and improving the quality of our work. Their willingness to listen, offer advice, and share their knowledge and experience have been invaluable. We are grateful for their commitment and dedication to helping us achieve our goals and realize our potential. Their mentorship has been a source of inspiration and motivation, and we are honored to have had the opportunity to work with them. Thank you for your invaluable contributions to our Final Year Project.

Abstract

Carpooling, also known as car-sharing, is a relatively new technology that has made travel more convenient and efficient for ordinary people. By sharing a vehicle and its associated costs, individuals can reduce their travel expenses, fuel consumption, and exhaustion. As the population grows, personal vehicles are no longer able to provide adequate transportation, leading to traffic congestion and increased air and noise pollution. Carpooling is the ideal solution to address these issues.

To facilitate carpooling, we plan to develop an Android-based application that will allow passengers to collaborate with like-minded people and plan their journeys using a user-friendly interface. Our primary goal is to offer a mobility service that enables users to plan, book, and pay for multiple transportation services through carpooling via a joint digital platform. This concept represents a shift away from personally owned modes of transportation and towards mobility provided as a service. The focus is on providing real-time solutions to travelers based on their specific travel needs. This carpooling system demonstrates how users can reach their destination by combining multiple rides.

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Introduction

This Carpooling system aims to provide one stop solution to daily commute problem without any hassle, making it more flexible and reliable while creating great connections with the help of an e-platform where commuters come together to share pleasant journey. Carpooling is a revolutionary and a unique way of travelling. We want to provide a hassle-free transit to our users along with a great experience.

TITLE : Eventive CarPooling

Illustration:



Supervisor : Nouman Durrani

Team : Muhammad Sameer Khan, M Faiz, Shah Hussain.

Roll Numbers : K191484, K190247, K191446

Spine : BS(CS) | Eventive Carpooling | 2023

Blurb : Sharing Rides , Connecting Communities

Quote : Collaborate with Commuters, Cut Your Commuting Costs and Reduce Your Carbon Footprint

Submitted on : May 18th, 2023

Department : Computer Science

Related Work

Carpoolyn [1] is Pakistan's 1st Carpooling (Ride-Sharing) App with an aim to make Traveling accessible, affordable, and safe for Everyone Across Pakistan. It connects the drivers with passengers traveling on the same routes to share the fuel cost. Have A Car? Just charge for empty seats in your car and make money to cover your commuting cost. Need A Ride? Share a ride with drivers heading towards the same destinations & reduce your traveling cost.

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FolksVagn [5] offers a community-based system that helps people share rides with others. While the passengers get rides at costs much cheaper than a taxi



service, the car owner gets a share of the fare. It is open only to corporate clients as it requires a corporate email for registration and has a prepaid account or online wallet system to pay for the ride.

Requirements

Hardware Interfaces

The application will be hosted. API will be used for location/gps functionality. The mobile application for User and Client can be installed on any mobile device which runs the latest Android versions or latest IOS version.

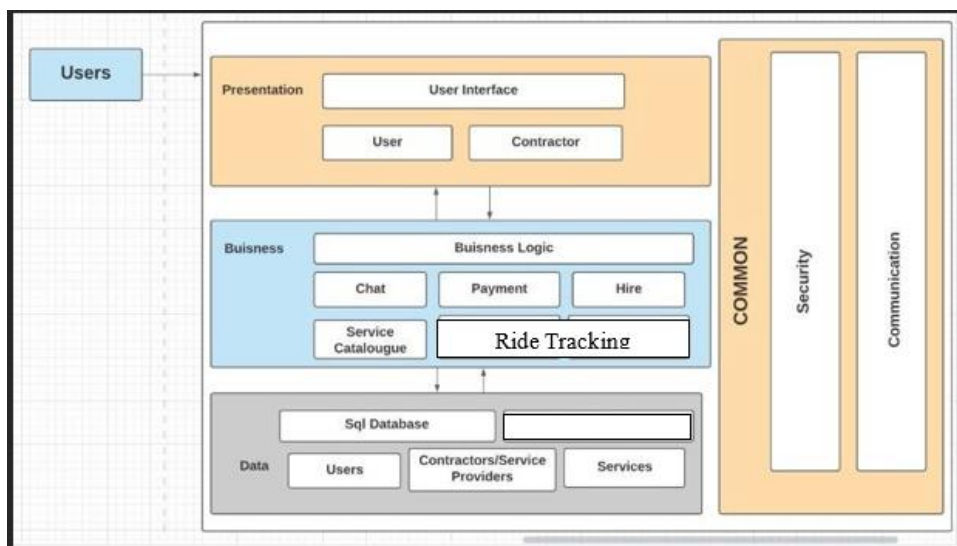
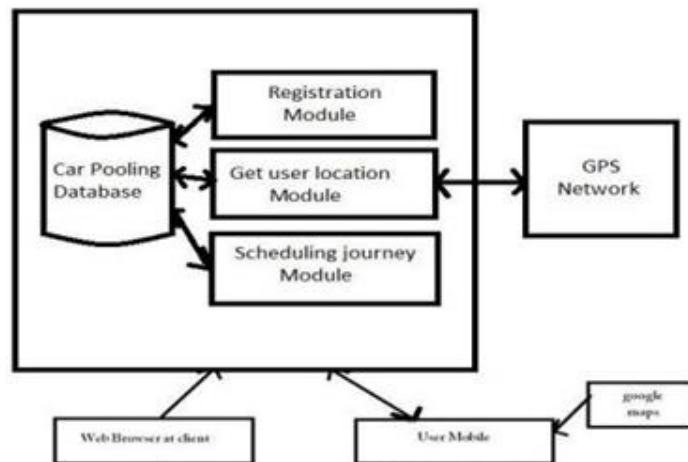
Software Interfaces

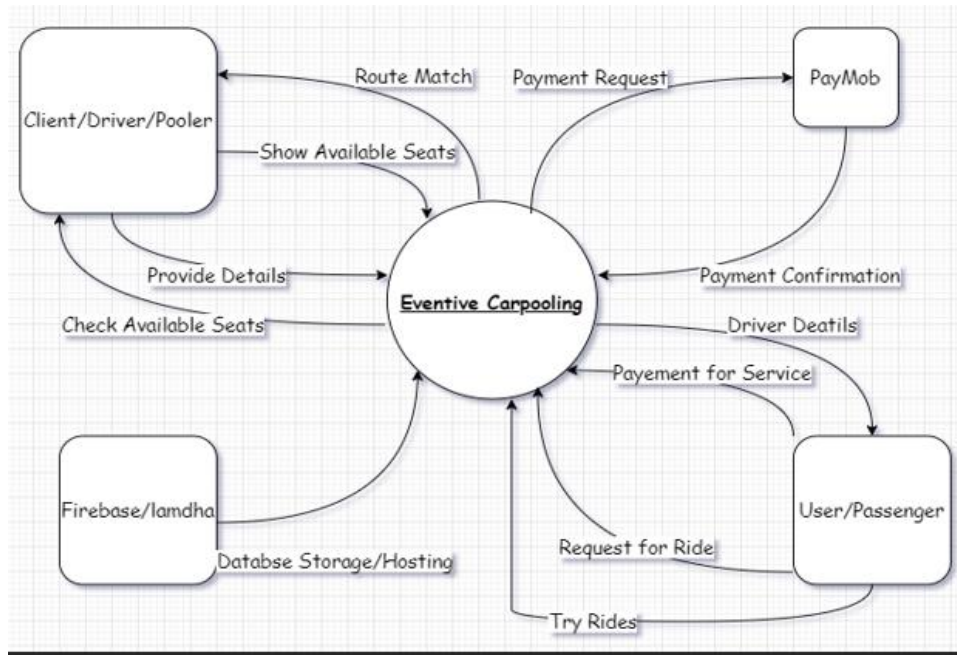
Application is developed using React Native. Firebase is used for database management. Through online payment feature is provided. Web application of Admin is not an operating system restricted as any system with browsers such as Chrome, Edge or Firefox can run the application. The mobile application of User and Client can be installed on the latest versions of Android or IOS.

Communications Interfaces

App will communicate with the user by showing possible rides and request rides. Payment details will be encrypted and secured. Password will be encrypted with Firebase Authentication which uses an internally modified version of script to hash account passwords. Even when an account is uploaded with a password using a different algorithm.

Designs





Tables

	Chat	Intra-City Sharing	Inter-City Sharing	Gender Specific	Realtime Connectivity	Road Statistics based fare
Savaree	✓	✓	✗	✗	✗	✗
Ridely	✓	✗	✓	✗	✗	✗
Carpoolyn	✓	✗	✗	✗	✗	✗
Careem	✓	✗	✗	✗	✗	✗
Uber	✓	✗	✗	✗	✗	✗
InDrive	✓	✗	✗	✗	✗	✗
Eventive	✓	✓	✓	✓	✓	✓

Table 1 : Comparative Analysis Table

Implementation

Carpooling system provides the benefits of sharing rides with different peoples. By surveying different papers, it is found that it uses heuristic searching techniques for searching the cars. By using Euclidean distance techniques, it finds the nearest car that has to be assigned to the passenger. It usually uses Global Positioning System (GPS) for finding the locations of cars. In study it is found that, Route matching

techniques are used in order to match drivers and passenger's locations by using Google maps API, and Then based on their route, the driver is allocated to the passenger. In most carpooling systems, it is observed that the Dijkstra algorithm is used for detecting the shortest route of destination for a passenger. The Dijkstra algorithm is usually used for detecting the optimal minimum path. It is found that it checks for maximum possibilities and then chooses the minimum route for the journey. Similarly, Dijkstra algorithm is also used to connect to the nearest vehicle.

Testing and Evaluation

Testing and evaluation are critical steps in ensuring the functionality, usability, and user experience of a carpooling app. Here are some steps for testing and evaluation:

1. **Functionality testing:** This testing ensures that the app is functioning correctly and without errors. Testers should try different features and scenarios to verify that they work as intended. For example, testers can check if the app can match drivers and passengers, book rides, and provide directions.
2. **Usability testing:** This testing focuses on the ease of use and user experience of the app. Testers should evaluate the app's user interface, navigation, and how easy it is for users to accomplish their goals. For example, testers can evaluate the clarity of the app's language, the placement of buttons, and how easy it is to create and join rides.
3. **Compatibility testing:** This testing ensures that the app works seamlessly across different devices and platforms. Testers should try the app on different devices, including smartphones, tablets, and desktops, with different operating systems and browsers. For example, testers can check if the app is compatible with iOS and Android devices and works on different screen sizes.
4. **Performance testing:** This testing evaluates the app's performance under different conditions. Testers should check how quickly the app loads and responds, how well it performs under different network conditions, and how well it handles large amounts of data. For example, testers can evaluate the app's response time, its ability to handle multiple users simultaneously, and its resilience under network interruptions.
5. **Security testing:** This testing ensures that the app is secure and protects user data. Testers should verify that the app uses secure protocols for data transfer and storage, and that it has appropriate authentication and access controls. For example, testers can evaluate if the app encrypts user data, if it has strong password policies, and if it protects against SQL injection attacks.
6. **User feedback and evaluation:** This evaluation involves getting feedback from real users of the app. Testers should conduct surveys or user interviews to gather feedback on the app's functionality, usability, and user experience. This feedback can then be used to identify areas for improvement and make changes to the app.

In summary, testing and evaluation are critical steps in ensuring the



functionality, usability, and user experience of a carpooling app. By conducting thorough testing and evaluation, developers can identify issues and improve the app's performance, security, and user experience.

Conclusion

In conclusion, a carpooling app can provide numerous benefits, including reducing traffic congestion, lowering transportation costs, and reducing carbon emissions. By leveraging technology, carpooling apps can connect drivers and passengers, making it easier to find rides and reduce the number of single-occupancy vehicles on the road. Additionally, carpooling apps can create a sense of community and help users build social connections.

To ensure the success of a carpooling app, it is essential to focus on user experience, functionality, and security. Thorough testing and evaluation can help identify areas for improvement and ensure that the app meets the needs of its users.

Overall, carpooling apps have the potential to create a more sustainable, cost-effective, and enjoyable transportation experience for commuters. As more people become aware of the benefits of carpooling, we can expect to see an increase in the adoption of carpooling apps and a reduction in traffic congestion, air pollution, and transportation costs.

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